FIZICĂ

$$\ddot{x} + 2\delta\dot{x} + \omega_0^2 x = 0$$

$$\downarrow$$

$$e^{\lambda t}(\lambda^2 + 2\delta\lambda + \omega_0^2) = 0$$

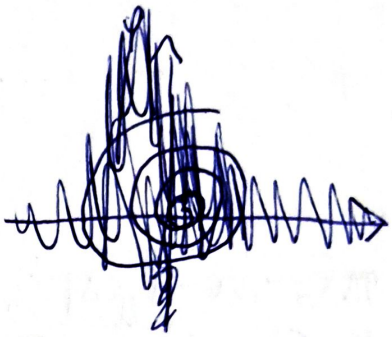
$$\downarrow$$

$$\lambda_{1,2} = -\delta \pm \sqrt{\delta^2 - \omega_0^2} \qquad \lambda_{1,2} = -\delta \pm i\omega_1$$

a) $\delta^2 < \omega_0^2 : \omega_1 = \sqrt{\omega_0^2 - \delta^2}$

$$x(t) = e^{-\delta t}(Ae^{i\omega_1 t} + Be^{-i\omega_1 t})$$

$$x(t) = Ae^{-\delta t}\sin(\omega_1 t + \varphi_1)$$

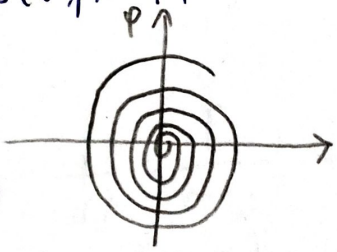


$$E = \frac{m\dot{x}^2}{2} + \frac{m\omega_0^2 x^2}{2}$$

$$\downarrow$$

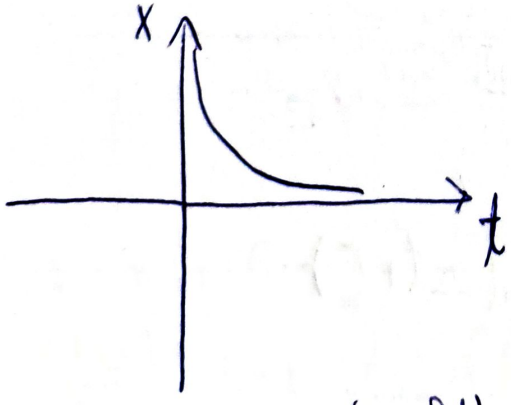
$$\cong \omega_0^2 \frac{m}{2}(Ae^{-\delta t})^2[\cos^2(\omega_1 t + \varphi_1) + \sin^2(\omega_1 t + \varphi_1)] \cong \frac{1}{2}K_{el}(Ae^{-2\delta t})$$

$$\boxed{\omega_1 \cong \omega_0}$$



b) $\delta^2 > \omega_0^2$

$$x(t) = e^{-\delta t}(Ae^{\omega_1 t} + Be^{-\omega_1 t}) = Ae^{-(\delta+\omega_1)t} + Be^{-(\delta-\omega_1)t}$$



$\nexists$ soluții

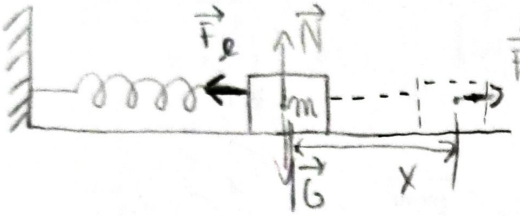
c) $\delta^2 = \omega_0^2 \; ; \; \exists \; (te^{\lambda t})$

$$x(t) = (A + Bt)e^{-\delta t}$$

Curs 3

Oscilații forțate

1. Ec. dif. a mișcării. Soluții
2. Fenomenul de rezonanță
3. Analiza electro-mecanică
4. Forțarea armonică



$\vec{F}_{ext} = \vec{F}_0 \cos(\Omega t)$

Ω ↓ pulsația (frecvența) forței ext

$$m\ddot{x} = -r\dot{x} - K_{el}x + F_0\cos(\Omega t)$$

$$\ddot{x} + \frac{r}{m}\dot{x} + \frac{K_{el}}{m}x = \frac{F_0}{m}\cos(\Omega t)$$

$\frac{r}{m}$ ↓ not. 2δ ; $\frac{K_{el}}{m}$ ↓ not. ω_0^2 ; $\frac{F_0}{m}$ ↓ not. f

$$\boxed{\ddot{x} + 2\delta\dot{x} + \omega_0^2 x = f e^{i\Omega t}} \quad (*)$$

↓

ec. dif. neomogenă de ordin 2 liniară

$x = x_{omogen} + x_{part.} \cong x_{part.}$ $\xrightarrow{\text{alegem}}$ $\tilde{C}e^{i\Omega t}$

x_{omogen} ↓ osc. amortizată

$\tilde{C}e^{i\Omega t}$ ↓ $Ce^{i\Phi}$ ↓ $C\cos(\Omega t + \Phi)$

$$\dot{x} = i\Omega \tilde{C} e^{i\Omega t}$$
$$\ddot{x} = (i\Omega)^2 \tilde{C} e^{i\Omega t}$$

$$\downarrow$$

$$(*) \Rightarrow (-\Omega^2 + i\Omega 2\delta + \omega_0^2)\tilde{C} e^{i\Omega t} = f e^{i\Omega t}$$

$$\Rightarrow \tilde{C} = \frac{f}{\omega_0^2 - \Omega^2 + i(2\Omega\delta)} = C e^{i\Phi} \Rightarrow$$

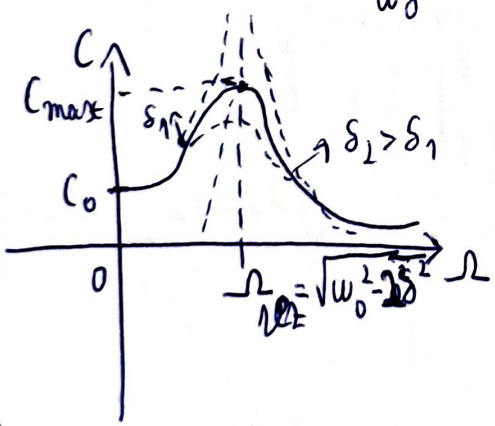
$$\tilde{C} = \frac{f(\omega_0^2 - \Omega^2)}{(\omega_0^2 - \Omega^2)^2 + (2\delta\Omega)^2} - i\frac{2\delta\Omega f}{(\omega_0^2 - \Omega^2)^2 + (2\delta\Omega)^2}$$

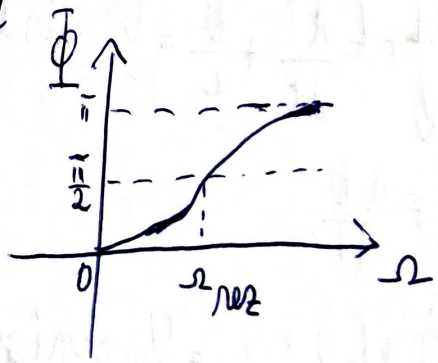
$$C = \frac{f}{\sqrt{(\omega_0^2 - \Omega^2)^2 + (2\delta\Omega)^2}}$$

$$\operatorname{tg}\Phi = -\frac{2\delta\Omega}{\omega_0^2 - \Omega^2}$$

Cazuri particulare

a) $\Omega = 0$; $(**) \Rightarrow C_0 = \frac{f}{\omega_0^2} = \frac{F}{m\omega_0^2} = \frac{F}{k_{el}}$





b) $\Omega = \infty \Rightarrow C \to 0$

c) $\left(\frac{dC}{d\Omega}\right)_{\Omega=\Omega_{rez}} = 0 \Rightarrow \frac{d}{d\Omega}\left[(\ldots)^{-1/2}\right] = -\frac{f}{2}(\ldots)^{-3/2}\cdot\left[2(\omega_0^2 - \Omega^2)(-2\Omega) + 2(2\delta\Omega)(2\delta)\right]\Big|_{\Omega_{rez}} = 0$

$\downarrow$

$\Omega_{rez} = \omega_0$

d) $\Omega_{rez} = \omega_0 \Rightarrow C_{max} \to \infty$!!

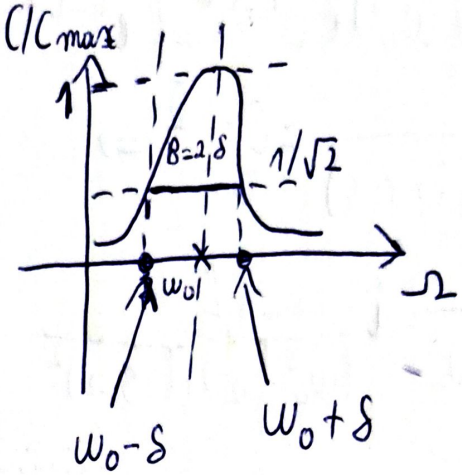
Amplitudine normată

$$\frac{C}{C_{max}} =$$

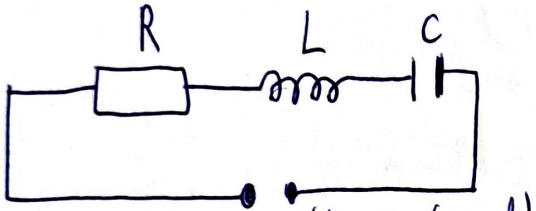
$$\downarrow$$

$$Q \cong \frac{\omega_0}{2\delta} = \frac{\omega_0}{B}$$

$B \searrow \Rightarrow Q \nearrow$



$\boxed{B = 2\delta}$



$u = U_0 \sin(\Omega t)$

$$u_R + u_L + u_C = u$$

$$RI + L\frac{dI}{dt} + \frac{Q}{C} = U_0 \sin(\Omega t)$$

$$\downarrow$$

$$I = \frac{dQ}{dt}$$

$$R\dot{Q} + L\ddot{Q} + \frac{Q}{C} = U_0 \sin(\Omega t)$$

$$\ddot{Q} + \frac{R}{L}\dot{Q} + \frac{1}{LC}Q = \frac{U_0}{L}\sin(\Omega t)$$

$$\downarrow \qquad \downarrow$$

$$2\delta = \frac{r}{m} \qquad \omega_0^2 = \frac{k_{el}}{m}$$

Mărimi mecanice	Mărimi electrice
x	Q
r (coef. de frec.)	R
m	L
K_{el}	$\frac{1}{C}$

$$\Rightarrow Q(t) = Q \sin(\omega t + \Phi)$$

$$\downarrow$$

$$I(t) = \dot{Q} = \frac{dQ}{dt} = \frac{U_0}{\sqrt{(X_L - X_C)^2 + X_R^2}}$$

$$\swarrow \qquad \searrow$$

$$L\omega \qquad \frac{1}{C\omega}$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t)$$

$$\downarrow$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega t)\, dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega t)\, dt$$

$$f(t) = \sum_n e^{-i n\omega t} C_n$$